

Unified Notation and Theorem Dependency Graph

This document consolidates notation across all three SCX theorems and maps the logical dependencies between them. It serves as a quick-reference companion to the polished theorem statements.

1 Unified Notation Table

1.1 Core Objects

Symbol	Meaning	Thm 1	Thm 2	Thm 3	Domain
$\mathcal{X}$	Input space	x	x	x	Measurable space
$\mathcal{Y}$	Label space, $K = \ \mathcal{Y}\ $	x	x	x	Discrete (classif.) or $\mathbb{R}^d$ (regr.)
X	Input random variable	x	x	x	$\mathcal{X}$
Y	Observed label	x	x	x	$\mathcal{Y}$
Y^*	True (latent) label	—	—	x	$\mathcal{Y}$

Symbol	Meaning	Thm 1	Thm 2	Thm 3	Domain
$f^* : \mathcal{X} \rightarrow \mathcal{Y}$	Ground-truth oracle	x	—	x	—
$\{f_m\}_{m=1}^M$	Expert models	x	x	x	$\mathcal{X} \rightarrow \mathcal{Y}$
$\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow [0, B]$	Bounded loss function	x	x	—	$B < \infty$
$\tau > 0$	Expert error threshold	x	—	—	$\mathbb{R}^+$

1.2 Noise Model

Symbol	Meaning	Thm 1	Thm 2	Thm 3	Definition
η	Global noise rate $\mathbb{P}(Y \neq Y^*)$	x	x	x	$\eta \in (0, 1/2)$
$\eta(x)$	Input-dependent noise rate	—	—	x	$\eta(x) = \mathbb{P}(Y \neq Y^* \mid X = x)$
Z	Noise indicator (1 = noise, 0 = clean)	—	x	—	$Z \in \{0, 1\}$
$e_m(x, y)$	Expert error indicator	x	—	—	$e_m = \mathbf{1}\{\ell(f_m(x), y) > \tau\}$

Symbol	Meaning	Thm 1	Thm 2	Thm 3	Definition
$C(x)$	Consistency score	x	—	—	$C(x) = \frac{1}{M} \sum_m e_m(x, y)$

1.3 State and Feature Structure

Symbol	Meaning	Thm 1	Thm 2	Thm 3	Definition
$\mathcal{S}$	State space	x	x	x	Finite set, $\ \mathcal{S}\ = K_S$
$s : \mathcal{X} \rightarrow \mathcal{S}$	State assignment	x	x	x	Partition of $\mathcal{X}$
$\rho_s = \mathbb{P}(X \in s)$	State probability	x	—	x	$\sum_s \rho_s = 1$
μ_s	State-level clean error bound	x	—	—	$\mu_s = \sup_{x \in s} \mathbb{E}[C \mid \text{clean}, x]$
C_{bal}	Error balance constant	x	—	—	$C_{\text{bal}} \geq 1$
Δ_s	Separation gap for state s	x	—	—	$\Delta_s = \min(\theta - \mu_s, 1 - C_{\text{bal}}\mu_s / (K - 1) - \theta)$
$\phi : \mathcal{X} \rightarrow \Phi$	Observed feature map	—	x	—	$\Phi \subseteq \mathbb{R}^{d_\phi}$

Symbol	Meaning	Thm 1	Thm 2	Thm 3	Definition
δ	Feature-state mutual information	—	x	—	$\delta = I(\phi(X); S)$
ε_ϕ	Normalized weakness	—	x	—	$\varepsilon_\phi = \delta / \log K_S \in [0, 1]$

1.4 Detection and Evaluation

Symbol	Meaning	Thm 1	Thm 2	Thm 3	Definition
θ	Detection threshold	x	—	—	$\theta \in (0, 1)$
$NS(x)$	Noise score	—	x	—	SCX noise score for sample x
$\hat{z}(x)$	Predicted noise label	—	x	—	$\hat{z}(x) = \mathbf{1}\{NS(x) > t\}$
FPR_s	False positive rate (state s)	x	—	—	$\mathbb{P}(C > \theta \mid \text{clean}, s)$
TPR_s	True positive rate (state s)	x	—	—	$\mathbb{P}(C > \theta \mid \text{noise}, s)$

Symbol	Meaning	Thm 1	Thm 2	Thm 3	Definition
F1	F1 score of detector	x	x	—	$2TP/(2TP + FP + FN)$
AUC	Area under ROC curve	—	x	—	$\mathbb{P}(\text{score}_{\text{noise}} > \text{score}_{\text{clean}})$
PR-AUC	Area under precision-recall curve	—	x	—	$\int \text{Precision}(t) d\text{Recall}(t)$

1.5 Theorem-Specific Parameters

Symbol	Meaning	Appears In	Defined As
$\mathcal{P}_{\text{noise}}, \mathcal{P}_{\text{hard}}$	Two data-generating processes	Thm 3	Constructive counterexample
$\varepsilon_1, \varepsilon_2$	Expert error rates in states s_1, s_2	Thm 3	$\varepsilon_1 \in (0, 1/2),$ $\varepsilon_2 \in (\varepsilon_1, 1/2]$
$\tilde{P}$	Auxiliary distribution $(\phi \perp S)$	Thm 2	$\tilde{P}(\phi, S) = P(\phi)P(S)$
$F1_{\text{base}}$	Loss-baseline detector F1	Thm 2	$\max_t F1(\hat{z}_{\text{loss}}(t))$
$F1_{\text{rand}}$	Random detector F1	Thm 2	$2\eta/(1 + \eta)$

2 Assumptions Cross-Reference

ID	Assumption	Thm 1	Thm 2	Thm 3
A1	Disjoint training sets	Required	Implicit	Target of Corollary 2
A2	Clean-data conditional independence	Required	Implicit	Technical basis for Cor. 2
A3	Bounded loss	Required	Implicit	—
A4	Uniform independent noise	Required	—	Target of Corollary 3
A5	State homogeneity	Required	—	Target of Corollary 4
A6	Balanced error distribution	Required	—	Target of Corollary 5

3 Theorem Dependency Graph

Theorem 3
(Unidentifiability)

Core claim:
Without A1-A6,
noise vs. difficulty
is observationally

equivalent

"What minimum assumptions
are needed to make the
problem tractable?"

v

A1-A6 are the minimal sufficient
conditions to break Thm 3's
unidentifiability

v

v

Theorem 1
(Noise Detection)

Positive result:
Under A1-A6, noise is
detectable with $F1 \rightarrow 1$
exponentially fast in M

Theorem 2
(Weak Feature Failure)

Negative result:
Even under A1-A6, if
features are ϵ -weak,
SCX cannot beat loss
baseline by more than
 $O(\sqrt{\epsilon})$

v

Unified SCX Picture

SCX Effectiveness

$$I(\cdot; S) = O(1/\sqrt{n})$$

- Thm 3: Why we need A1-A6
- Thm 1: What we gain with them
- Thm 2: Where the boundary lies

3.1 Dependency Logic

The three theorems form a logical chain of necessity, sufficiency, and boundary:

1. **Theorem 3 (Fundamental Hardness)** establishes that without structural assumptions, the noise detection problem is ill-posed: noise and intrinsic difficulty are observationally equivalent. This is the **necessity** result – it explains *why* assumptions are required, not just convenient.
2. **Theorem 1 (Sufficient Conditions)** shows that under assumptions A1-A6 – the minimal set identified in Theorem 3 – noise detection succeeds with exponentially decaying error. This is the **sufficiency**

result – it quantifies *how well* SCX works when its assumptions hold.

3. **Theorem 2 (Boundary Condition)** shows that even when A1-A6 hold, the feature representation ϕ must carry enough mutual information about the true state S for SCX to outperform a simple loss baseline. This is the **boundary** result – it identifies *when* SCX ceases to add value despite valid assumptions.

3.2 Information-Theoretic Unification

The three theorems can be unified under a single information-theoretic inequality:

$$\underbrace{\text{SCX Advantage}}_{\text{Thm 1: exponential F1} \rightarrow 1} \leq \underbrace{f(I(\phi; S))}_{\text{Thm 2: } O(\sqrt{\delta}) \text{ bound}} \leq \underbrace{g(\text{A1-A6})}_{\text{Thm 3: minimal conditions}}$$

where f and g are monotone increasing functions. The chain reads: - Without A1-A6, $g = 0$ and the problem is unidentifiable (Thm 3) - With A1-A6 but $I(\phi; S) \rightarrow 0$, SCX collapses to the baseline (Thm 2) - With both A1-A6 and strong features, SCX achieves exponential guarantee (Thm 1)

4 Key Inequalities Reference

Inequality	Used In	Statement
Hoeffding	Thm 1 (Lemmas 2, 3)	$\mathbb{P}(\bar{X} - \mu \geq t) \leq e^{-2nt^2}$
Chernoff (Sanov)	Thm 1 (tightened)	$\mathbb{P}(\bar{X} \geq \theta) \leq e^{-n \cdot \text{KL}(\theta \ \mu)}$

Inequality	Used In	Statement
Pinsker	Thm 2 (original)	$TV(P, Q) \leq \sqrt{\frac{1}{2} KL(P\ Q)}$
Bretagnolle-Huber	Thm 2 (tightened)	$TV(P, Q) \leq \sqrt{1 - e^{-KL(P\ Q)}}$
Fano	Thm 2 (Lemma 1)	$P(\hat{S} \neq S) \geq \frac{H(S \phi) - \log 2}{\log \ \mathcal{S}\ }$
Data Processing	Thm 2	$TV(P_{\text{pred}}, Q_{\text{pred}}) \leq TV(P, Q)$

5 Quick-Start Guide

If you want to understand...	Read this combination
Why SCX needs assumptions	Thm 3 + Section 4 of Thm 3 (A1-A6 mapping)
How well SCX detects noise	Thm 1 + Corollaries 1-4
When SCX cannot work	Thm 2 + Diagnostic Section 7
Practical deployment	Thm 1 Corollary 4 (finite-sample) + Thm 2 diagnostics
The complete picture	All three + this notation document